Compulsory Exercise

Topics in Life Insurance (Liv2), Block 3, 2026

March 15, 2026

Contents

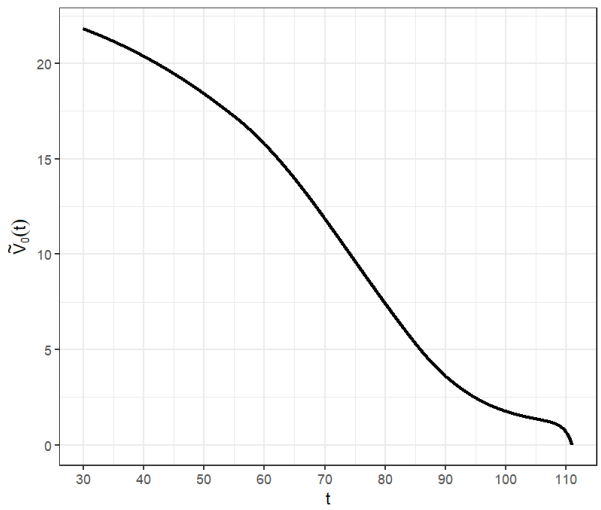
(i)

Using Lebniz' rule we obtain af ordinary backward differential equation for the reserve $\tilde{V}_0$. We have

$$\begin{aligned}
\frac{d}{dt}\tilde{V}_0(t) &= \int_t^\infty \frac{\partial}{\partial t} e^{-\int_t^s (\tilde{r}(u)+\tilde{\mu}(u))\,du}\,ds - 1 \\
&= \int_t^\infty e^{-\int_t^s (\tilde{r}(w)+\tilde{\mu}(w))\,dw}(\tilde{r}(t)+\tilde{\mu}(t))\,ds - 1 \\
&= (\tilde{r}(t)+\tilde{\mu}(t)) \int_t^\infty e^{-\int_t^s (\tilde{r}(w)+\tilde{\mu}(w))\,dw}\,ds - 1 \\
&= (\tilde{r}(t)+\tilde{\mu}(t))\tilde{V}_0(t) - 1,
\end{aligned}$$

with $\tilde{V}_0(111) = 0$.

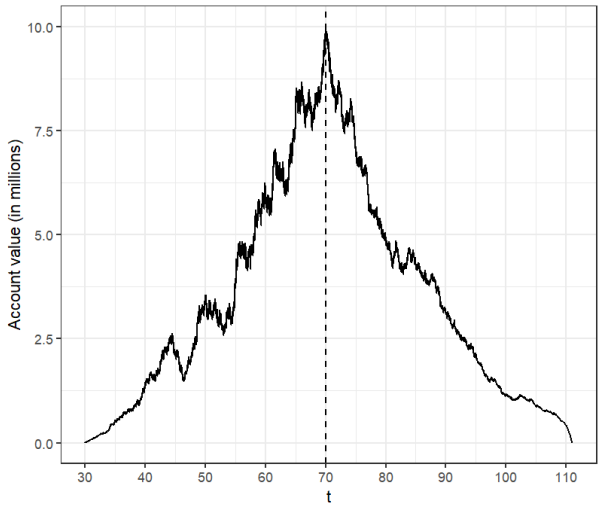
Now, using the fourth order Runge-Kutta method we optain a numerical solution of $\tilde{V}_0(t)$. The result is shown via the plot below



Exercise (ii)

(iii)

A simulation of the account is provided below.



We observe from our simulation that it begins from age $t = 30$ at the value of zero as demanded. Thereafter, the account accumulates until retirement $m = 70$ with a value of around 9.8 million (monetary units). Then the account de-cumulates until our assumed lifetime cut-off at $t = 111$. We also observe random fluctuates due to the fact that we model the risky asset via a Brownian motion.

Exercise (iv)

We argue that $V^0(t,x) = x$ by verification: we show that $V^0(t,x) = x$ satisfies Thiele's partial differential equation (PDE) in both phases with the correct terminal boundary condition, and conclude by uniqueness.

Thiele's PDE

The market reserve $V^0(t,x)$ is defined as the expected discounted future net outflows,

$$V^0(t,x) = \mathbb{E}^{\mathbb{P}\otimes\mathbb{Q}}\left[\int_t^{111} e^{-\int_t^s (\tilde{r}(v)+\mu(v))\,dv}\, c(s,X(s))\,ds \;\middle|\; X(t)=x, Z(t)=0\right].$$

By the Feynman–Kac theorem—equivalently, by applying Itô's formula to the discounted process $e^{-\int_0^t (\tilde{r}+\mu)\,ds} V^0(t,X(t))$ and requiring its drift to equal the outflow rate—$V^0(t,x)$ satisfies Thiele's PDE

$$\frac{\partial V^0}{\partial t} + a(t,x)\,\frac{\partial V^0}{\partial x} + \tfrac{1}{2}\,b(t,x)^2\,\frac{\partial^2 V^0}{\partial x^2} = \big(\tilde{r}(t)+\mu(t)\big)V^0 - c(t,x), \tag{1}$$

where $a(t,x)$ and $b(t,x)$ are the drift and diffusion of X, and $c(t,x)$ is the net outflow rate (benefits minus premiums).

Derivatives of $V^0(t,x) = x$

We conjecture $V^0(t,x) = x$ and compute its partial derivatives:

$$\frac{\partial V^0}{\partial t} = 0, \qquad \frac{\partial V^0}{\partial x} = 1, \qquad \frac{\partial^2 V^0}{\partial x^2} = 0.$$

Substituting into (1), the PDE reduces to verifying

$$a(t,x) = \big(\tilde{r}(t)+\mu(t)\big)x - c(t,x). \tag{2}$$

Retirement phase, $t \in [m, 111]$

From SDE (2), the drift coefficient is

$$a(t,x) = \big(\tilde{r}(t)+\mu(t)\big)x - \frac{x}{\tilde{V}^0(t)},$$

There is no death benefit during retirement (the account is released to the pool on death), so the net outflow rate consists solely of the annuity payout, $c(t,x) = x/\tilde{V}^0(t)$.

We need to verify (2), i.e. that $a(t,x) = \big(\tilde{r}+\mu\big)x - c(t,x)$. Equivalently, we show that the left-hand side minus the right-hand side is zero:

$$\begin{aligned}
&a(t,x) - \big(\tilde{r}+\mu\big)x + c(t,x)\\
&= \left[\big(\tilde{r}+\mu\big)x - \frac{x}{\tilde{V}^0(t)}\right] - \big(\tilde{r}+\mu\big)x + \frac{x}{\tilde{V}^0(t)}\\
&= 0. \quad \checkmark
\end{aligned}$$

Active phase, $t \in [30, m)$

From SDE (1), the drift is

$$a(t,x) = \tilde{r}(t)\,x + \theta L(t) - \mu(t)\big(KL(t) - x\big)$$

During the active phase the policyholder pays premiums at rate $\theta L(t)$ (an inflow to the insurer) and the insurer pays death benefits at expected rate $\mu(t)\,KL(t)$. The net outflow rate is therefore

$$c(t,x) = \mu(t)\,KL(t) - \theta L(t).$$

We verify (2) in the same way as above:

$$\begin{aligned}
&a(t,x) - \big(\tilde{r} + \mu\big)x + c(t,x) \\
&= \big[(\tilde{r} + \mu)x + \theta L - \mu KL\big] - (\tilde{r} + \mu)x + \big[\mu KL - \theta L\big] \\
&= 0. \quad \checkmark
\end{aligned}$$

Boundary condition

At the terminal age $t = 111$ there are no future obligations, so the boundary condition is $V^0(111, x) = 0$. If $V^0(t,x) = x$, this requires $X(111) = 0$. This is guaranteed by construction of the calculation basis: the reserve $\tilde{V}^0(t)$ computed in Exercise (i) satisfies $\tilde{V}^0(111) = 0$, so the annuity payout rate $X(t)/\tilde{V}^0(t) \to \infty$ as $t \to 111$, which drains the account completely. In other words, the purpose of $\tilde{V}^0$ is precisely to calibrate the payout rate so that the account is exhausted at the maximum age. The simulation from Exercise (ii) confirms this numerically: $X(111) \approx 0$.

Conclusion

Since $V^0(t,x) = x$ satisfies Thiele's PDE (1) in both phases and matches the terminal boundary condition, uniqueness of the solution gives

$$\boxed{V^0(t,x) = x.}$$

The market reserve equals the account value. This is the defining property of a unit-link product: the policyholder bears all investment risk, and the account dynamics are constructed to continuously budget for every mortality and benefit cash flow, making the account the exact self-financing replicating portfolio for all future obligations.

Exercise (v)

Exercise (vi)

Exercise (vii)